



DPUSNDXTRF

1. Inputs:

<i>ignoreTRFid</i>	TRF-ID which is not simulated default: 0 (ego-vehicle)
<i>GlobalVolume</i>	global volume of the sound sources [0..1]
<i>EngineVolume</i>	volume for the engine sound sources
<i>WindRollVolume</i>	volume for wind and roll sound sources
<i>MiscVolume</i>	volume for horn, siren and misc sound sources
<i>MiscTRFid1</i>	vehicles with the given TRF-ID play the custom
<i>MiscTRFid2</i>	(misc) sound file
<i>MiscTRFid3</i>	default: -1 (none played)
<i>v1WR</i>	lowest speed [m/s] for wind and roll sounds default: 2.778
<i>v2WR</i>	speed [m/s] for full wind and roll sounds default: 8.333
<i>FakeRPM</i>	use fake RPM modell instead of the data from TRF
<i>FadeIn</i>	fade in new sound sources detected within the radius
<i>FadeInTime</i>	time [ms] for the volume to fade from 0 to max volume default: 500
<i>EnableLogOutput</i>	additional output written to SILABdebug.log default: 0

2. Parameters:

<i>NSources</i>	maximum number of simulated vehicles (A vehicle may have more than one source!) (Maximum number of available sound sources depends on the hardware.) default: 10
<i>Radius</i>	vehicle sounds are simulated within this radius [m] (If the number of vehicles is higher than NSources simulation of the vehicles closest to the listener is done.) default: 250
<i>BufferCount</i>	number of buffers for each sound source default: 10
<i>BufferLengthMS</i>	length of each buffer [ms] default: 16
<i>ReferenceDistance</i>	distance [m] before a sound source softens (according to the OpenAL sound model) default: 5
<i>RolloffFactor</i>	factor for the above mentioned softening. Values < 1 lead to slower softening, for values > 1 it is faster. default: 1.0



3. Outputs:

<i>NActive</i>	number of vehicles within the range for sound simulation
<i>NActiveSources</i>	number of sound sources currently in use

4. Functional principle:

DPUSNDXTRF was developed to simulate the sound of surrounding vehicles. Each vehicle may use up to three sound sources, divided into engine, wind/roll and misc.

The engine sound is simulated using a granular sound source, a sweep of a sample-led engine WAV file from idle to full RPM. Position within that file is either chosen by the RPM simulation of the TRF module or a fake RPM simulation (see below for details).

Wind and roll sound are done by a looping WAV file. It is possible to give a range of speed where the volume of the sound is adjusted (ramped in and out). Below that range, the volume will be zero, so no wind and roll sound is played for still standing or slow moving vehicles. Above that range, the maximum volume will be used. By default, the range is from 10km/h to 30km/h.

The misc slot is either used for horn, siren or a custom sound given by the configuration. To play the custom sound, the corresponding TRF-ID must be set as an input of MiscTRFid1 to MiscTRFid3.

Note: Maximum number of sound sources for a Realtek-Onboard-Soundcard is about 32. The Creative Soundblaster X-Fi can handle up to 128 sources. 256 Quellen, allerdings wird der Prozessor dadurch sehr stark belastet.

In any case one source is reserved for EAX as well as two other blocked sources for stereo playback. (Standard sound simulation SILAB)

5. Example for a driving noise configuration:

```
SILAB ASM
{
    GranularSound pkw
    {
        TRFType = {0,1};
        SubType = {0};
        EngineFile = "Granular/SWEEP_750-4500_CarEngine.wav";
        RollWindFile = "TRF/RollWind_Car.wav";
        HornFile = "TRF/Horn_Car.wav";
        SirenFile = "TRF/Siren_Car.wav";
        MiscFile = "diverse/blinker.wav"; # Dummy für DPUSNDXTRF
        Grains = 3;
        EQlow = 0.5;
        EQmiddle = 0.5;
        EQhigh = 0.5;
        RPM = 4500;
    };
};
```



```
};
```

The configuration consists of the following parts:

SILAB ASM marks this section as configuration for DPUSNDXTRF.

GranularSound declares a sound source which is designated for vehicle sounds done by a WAV file sweep. The *TRFType* list defines which vehicles can be re-presented by this configuration part (0 = ego-vehicle, 1 = passenger car, 2 = truck, 9 = motorbike). The additional parameter *SubType* may be used to define different configurations within one class, for example small passenger cars, SUVs, sport cars, muscle cars and so on.

EngineFile should point to the WAV file containing the sweep from idle to full RPM. In the example it is located under *SILAB/lib/sounds/Granular/* named *SWEEP_750-4500_CarEngine.wav* indicating a sweep from 750 RPM (idle) to 4500 RPM recorded from an Opel Zafira. Accordingly, the parameter *RPM* is set to 4500. The *GranularSound* source uses the set number of *Grains* and has an equalizer (*EQlow*, *EQmiddle*, *EQhigh*) to use one sweep with different settings for different car types. If the *RPM* value of the TRF module is used, the playback position [0..1] is calculated by $\frac{\text{value from TRF}}{\text{max RPM given}}$. Otherwise if *FakeRPM* is used, the base position is 0.1 (for $v \leq 0$ and $a \leq 0$). Higher speeds and accelerations add to that base position value up to 0.5 (for $v \geq 13\text{m/s}$) and up to 0.4 (for $a \geq 5\text{m/s}^2$).

RollWindFile is a WAV file containing a loopable sample of the wind and roll sound of a passing car.

HornFile is the (loopable) WAV file of a horn, *SirenFile* of a siren. *MiscFile* may be any (loopable) WAV file one wants to play back during simulation.